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INTEGRATED CEILING DEVICE WITH MECHANICAL ARRANGEMENT FOR A LIGHT SOURCE

Abstract

An integrated ceiling device includes integrated ceiling device including an electronic device housing, a heat dissipating structure, a light source, and a reflection/refraction assembly. An air gap is defined between the electronics housing and other components allowing ambient air to flow therethrough for cooling.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. patent application Ser. No. 18/619,237 filed Mar. 28, 2024, which is a continuation of U.S. patent application Ser. No. 18/238,521, filed Aug. 28, 2023 (now Abandoned), which is a continuation of U.S. patent application Ser. No. 18/213,306 filed Jun. 23, 2023 (now U.S. Pat. No. 11,944,053), which is a continuation of U.S. patent application Ser. No. 17/517,451, filed Nov. 2, 2021 (now U.S. Pat. No. 11,730,100), which is a continuation of U.S. patent application Ser. No. 17/142,114, filed Jan. 5, 2021 (now U.S. Pat. No. 11,172,627), which is a continuation of U.S. patent application Ser. No. 16/883,028, filed May 26, 2020 (now U.S. Pat. No. 10,941,783), which is a continuation of U.S. patent application Ser. No. 15/089,146, filed Apr. 1, 2016 (now U.S. Pat. No. 10,677,446), which is a continuation of U.S. patent application Ser. No. 14/151,245, filed Jan. 9, 2014 (now U.S. Pat. No. 9,441,634), which claims priority to U.S. Provisional Patent Application No. 61/751,660, filed Jan. 11, 2013. The disclosures of each of which are hereby incorporated herein by reference in their entireties.

TECHNICAL FIELD

[0002] The present invention relates generally to integrated ceiling device technology including lighting. More specifically, the present invention relates to an integrated ceiling device including a

mechanical arrangement for a light emitting diode (LED) light source having effective heat dissipation capability and efficient optics.

BACKGROUND

[0003] Historically, the building industry has employed a large number of professions to design, manufacture, and maintain building systems that perform a variety of functions. These various functions include, for example, lighting control, smoke detection, air quality monitoring, occupancy awareness, and so forth. Each individual system carries with it costs associated with upfront equipment purchase, installation, operation, and maintenance. While cost control is important, additional factors such as aesthetic appeal, ease of use and maintenance, expandability, and so forth can be equivalently critical in the design, manufacture, operation, and maintenance of a variety building systems.

[0004] Increasingly, industry is focusing on intelligent systems or smart systems to provide a variety of building system functions. Unfortunately, these intelligent systems can be costly, complex, and difficult to maintain. Moreover, due at least in part to historical legacy, few advances have been made in offering building owners efficient, economical, and aesthetically pleasing smart building solutions.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0005] A more complete understanding of the present invention may be derived by referring to the detailed description and claims when considered in connection with the Figures, wherein like reference numbers refer to similar items throughout the Figures, the Figures are not necessarily drawn to scale, and:

[0006] FIG. 1. shows a top perspective view of a mechanical arrangement for an integrated ceiling device, i.e., a LEAM, in accordance with an embodiment;

[0007] FIG. 2. shows a bottom perspective view of the mechanical arrangement of FIG. 1;

[0008] FIG. 3. shows a top view of the mechanical arrangement;

[0009] FIG. 4. shows a bottom view of the mechanical arrangement

[0010] FIG. 5. shows a side view of the mechanical arrangement

[0011] FIG. 6. shows a side sectional view of the mechanical arrangement

[0012] FIG. 7. shows a top perspective view of an integrated ceiling device, i.e., a LEAM, in accordance with another embodiment;

[0013] FIG. 8. shows a bottom perspective view of the LEAM of FIG. 7;

[0014] FIG. 9. shows a top view of the LEAM of FIG. 7;

[0015] FIG. 10. shows a bottom view of the LEAM of FIG. 7;

[0016] FIG. 11. shows a side view of the LEAM of FIG. 7;

[0017] FIG. 12. shows a side sectional view of the LEAM of FIG. 7;

[0018] FIG. 13. shows a top perspective view of an integrated ceiling device, i.e., a LEAM, in accordance with another embodiment;

[0019] FIG. 14. shows a bottom perspective view of the LEAM of FIG. 13;

[0020] FIG. 15 shows a top view of the LEAM of FIG. 13;

[0021] FIG. 16. shows a bottom view of the LEAM of FIG. 13;

[0022] FIG. 17. shows a side view of the LEAM of FIG. 13;

[0023] FIG. 18. shows a side sectional view of the LEAM of FIG. 13;

[0024] FIG. 19. shows a bottom perspective view of a device that may be mounted to the mechanical arrangement of FIG. 1 in accordance with another embodiment;

[0025] FIG. 20. shows a top perspective view of the device of FIG. 19;

[0026] FIG. 21. shows a side view of the device of FIG. 19;

[0027] FIG. 22. shows a bottom view of the device of FIG. 19;
[0028] FIG. 23. shows a top view of the device of FIG. 19;
[0029] FIG. 24. shows a bottom perspective view of a device that may be mounted to the mechanical arrangement of FIG. 1 in accordance with another embodiment;
[0030] FIG. 25. shows a top perspective view of the device of FIG. 24;
[0031] FIG. 26. shows a side view of the device of FIG. 24;
[0032] FIG. 27. shows a bottom view of the device of FIG. 24;
[0033] FIG. 28. shows a top view of the device of FIG. 24;
[0034] FIG. 29. shows a side view of the light fixture of FIG. 13 with the devices of FIGS. 19 and 24 retained on the light fixture;
[0035] FIG. 30 shows a block diagram of an electronics assembly including a variety of devices that may be included within the electronics assembly 56 for any of the LEAMs; and
[0036] FIG. 31A-31H show top views of different configurations of the LEAM of FIG. 7.

DETAILED DESCRIPTION

[0037] Suitable ambient lighting is a quintessential need in virtually every building system application, and the lighting industry is rapidly migrating from traditional light sources such as fluorescent, high intensity discharge (HID), and incandescent lamps to solid state lighting, such as light-emitting diodes (LEDs). Light fixtures (technically referred to as luminaires in accordance with International Electrotechnical Commission terminology) employing LEDs initially appeared in small devices utilized in low light output applications. Increasingly, light fixtures employing LEDs can be found in indoor commercial applications, such as predominantly high-end offices, institutional spaces, and supermarkets' refrigerated spaces. In exterior applications, municipalities and some large box retailers have begun replacing their traditional street and pole mounted light fixtures with fixtures employing LEDs. LED technology is also being embraced by the automotive and aircraft industries.

[0038] An LED lamp is a solid state device. The solid state technology can enable device integration in an un-paralleled manner thus leading to opportunities in the areas of efficient energy usage, efficient use of human resources, safer and more pleasant illumination, and better use of material resources. Indeed, light fixtures employing LEDs are fast emerging as a superior alternative to conventional light fixtures because of their low energy consumption, long operating life, optical efficiency, durability, lower operating costs, and so forth.

[0039] There are presently a number of technical and economic problems associated with the implementation of high-output LED light fixtures in the market. The LED lamp cost is high when compared with traditional light sources. Smaller LED lamps yield higher efficiency. However, to generate high light output with LED lamps, clusters of LED lamps need to be formed. The more LED lamps used, the higher the cost. Additionally, cool operation is essential to the electronics devices and particularly to the LED lamp.

[0040] A cluster of high output LED lamps in close proximity to one another generates a significant amount of heat. Thus, implementation of LEDs for many light fixture applications has been hindered by the amount of heat build-up. High temperature reduces the lamp efficiency and may shorten the life of the lamp and other electronic components, eventually causing device failure. Additionally, the life of the LED lamp and its output depends on the surrounding ambient temperature, and most critically, its impact on the lamps' junction temperature. The junction temperature is the temperature where the lamp die is secured to the factures' heat sink. As the heat generated with high output LED lamps increases, so does the difficulty of designing large passive heat sinks that are architecturally attractive, lightweight, and economically feasible. Consequently, effective heat dissipation is an important design consideration for maintaining light output and/or increasing lifespan. of the LED light source.

[0041] Embodiments within the present disclosure include an integrated ceiling device and a mechanical arrangement that provides effective heat dissipation for a number of light sources

installed in the integrated ceiling device. For brevity, the integrated ceiling device is referred to herein as a Local Environmental Area Manager (LEAM). The LEAM, with the mechanical arrangement, is configured to accommodate multiple LED light sources. The mechanical arrangement maintains low junction temperature by effectively conducting heat generated by the LED light sources, also referred to herein as LED lamps, away from other LED lamps and other electronic components. Maintaining a low temperature at this junction yields improvements in lamp energy efficiency and enhanced lifespan for the LED light sources.

[0042] Additionally, the configuration of the mechanical arrangement physically isolates the heat dissipating structure of the mechanical arrangement from a housing in which an electronics assembly for the LEAM is housed. As such, the housing may be sized to accommodate a plurality of onboard electronic devices (e.g., camera, occupancy sensor, air quality sensor, smoke detector, and so forth) that are not unduly taxed by the heat produced by the LED light sources. These onboard electronic devices may be configured to emulate human sensory capability and to make actionable decisions based on changing environmental conditions in which the LEAM is located. As such, the LEAM can be configured as a smart system to provide a variety of building system functions. Accordingly, the LEAM includes several elements that are organized in a manner that resolves the mechanical, thermal, electrical, and architectural challenges that are commonly associated with the design of high-output LED light fixtures and other ceiling mounted devices. Further the structural configuration of the LEAM makes the LEAM suitable for use in a wide variety of environments, such as, commercial, institutional, and industrial applications.

[0043] Referring now to FIGS. 1-6, FIG. 1 shows a top perspective view of a mechanical arrangement 20 for an integrated ceiling device, i.e., a LEAM in accordance with an embodiment. FIG. 2 shows a bottom perspective view of mechanical arrangement 20. FIG. 3 shows a top view of mechanical arrangement, FIG. 4 shows a bottom view of mechanical arrangement 20, FIG. 5 shows a side view of mechanical arrangement, and FIG. 6 shows a side sectional view of mechanical arrangement 20. The inclusion of mechanical arrangement 20 into various LEAM embodiments is discussed below in reference to FIGS. 7-18.

[0044] Mechanical arrangement 20 generally includes a housing 22, a heat dissipating structure 24, and support arms 26. Housing 22, heat dissipating structure 24, and support arms 26 may be monolithically casted or printed, or can be assembled by joining casted and non-casted elements. Heat dissipating structure 24, as well as housing 22 and support arms 26 may be manufactured from a heat dissipating, non-corrosive material and may be painted or otherwise treated to suit architectural needs. The configuration of mechanical arrangement provides a rigid design suitable in adverse environments, and housing 22, heat dissipating structure 24, and support arms 26 are organized in a manner that maximizes air flow across the elements.

[0045] With particular reference to FIGS. 3 and 5, housing 22 exhibits a generally cylindrical shape having an outer diameter 28 and a height 30 along a longitudinal axis 32 of mechanical arrangement 20 (see FIG. 5). In alternative architectural configurations, housing 22 need not be cylindrical in shape, but may instead be any other suitable three-dimensional shape. Additionally, housing 22 may be expanded both vertically and horizontally to accommodate device scalability.

[0046] Heat dissipating structure 24 includes a central opening 34 surrounded by a plurality of fins 36 having a height 38 (see FIG. 5). Thus, heat dissipating structure 24 is generally ring-shaped, with central opening 34 exhibiting an inner dimension, and more particularly, an inner diameter 40 (best seen in FIGS. 3-4). In the illustrated embodiment, heat dissipating structure 24 is a circular ring-shaped structure corresponding with the shape of housing 22. However, in alternative architectural configurations, heat dissipating structure 24 may have a different surrounding shape, e.g., rectangular, oblong, triangular, and so forth, while still retaining central opening 34. (see FIGS. 31A-31H.) It should be further understood that in alternative architectural configurations, central opening 34 need not be circular, but could instead have another shape corresponding to, or differing from, the shape of housing 22 and/or heat dissipating structure 24.

[0047] Housing 22 is positioned within central opening 34, and support arms 26 extend between and interconnect housing 22 with heat dissipating structure 24. Outer diameter 28 of housing 22 is less than inner diameter 40 of central opening 34. Therefore, housing 22 is physically spaced apart from fins 36 by an air gap 42 extending between housing 22 and heat dissipating structure 24. The configuration of fins 36 permits free air flow of no less than two hundred and seventy degrees across its vertical axis and the configuration of housing 22 permits no less than three hundred and twenty degrees of free air flow across its vertical axis and in between support arms 26. In addition, housing 22 and heat dissipating structure 24 are exposed to air at their tops and bottom faces. Thus, air is free to flow, as indicated by the arrows in FIG. 7, between housing 22 and heat dissipating structure 24 and over housing 22 and heat dissipating structure 24 to provide effective cooling. Furthermore, housing 22 and its contents are protected from moving objects in the surrounding area by fins 36 of heat dissipating structure 24.

[0048] Outer vertical walls 45 at each quarter section of heat dissipating structure 24 can include bores 47, i.e., a hole or passageway, which can serve as an attachment point for a decorative cover, protective frame, protective reflector frame, and the like (not shown). Additionally, the plurality of fins 36 are spaced about the circumference of heat dissipating structure 24. In particular, fins 36 are generally uniformly distributed about both an outer perimeter 44 and an inner perimeter 46 of heat dissipating structure 24. Thus, fins 36 extend partially into air gap 42 between housing 22 and heat dissipating structure 24. The multitude of fins 36 maximize airflow about heat dissipating structure 24 and thereby facilitate effective heat dissipation.

[0049] Heat dissipating structure 24 further includes a generally ring-shaped bottom face 48 (FIGS. 2, 4, and 5) connected with fins 36 and at least one lamp seat 50 formed in bottom face 48. Heat dissipating structure 24 and bottom face 48 may be formed as two separately manufactured components that are bolted, welded, or otherwise coupled together during manufacturing. In the illustrated an embodiment a plurality of lamp seats 50 are formed in bottom face 48 of heat dissipating structure 24, and are generally uniformly distributed in bottom face 48.

[0050] Bottom face 48 further includes recessed channels 52 (FIGS. 2, 4, 5) formed therein. Recessed channels 52 are suitably formed and routed to provide the locations in bottom face 48 for electrically interconnecting each of lamp seats 50. That is, wiring (not shown) may be directed through recessed channels 52 when mechanical arrangement 20 is assembled with other components (discussed below) to form a particular LEAM with lighting capacity. Recessed channels 52 are illustrated in FIGS. 2, 4, 5 for exemplary purposes. In actual practice, recessed channels 52 would not be visible on an exterior surface of bottom face 48 of heat dissipating structure 24.

[0051] Each lamp seat 50, in the form of, for example, a direct mounted die, is configured to receive a light source 54 (see FIG. 8). Light source 54 may be any suitable lamp or light array, such as an LED lamp. Each lamp seat 50 extends inwardly into heat dissipating structure 24 so that each lamp seat 50 is generally surrounded by fins 36. The configuration of heat dissipating structure 24 with fins 36 effectively conducts heat generated by the LED light sources 54 away from LED light sources 54. Maintaining a low temperature at lamp seats 50 yields improvements in lamp energy efficiency and enhanced lifespan for the LED light sources 54.

[0052] In its centralized location in central opening 34 of heat dissipating structure 24, housing 22 functions to centralize power distribution and serves as a data receiving and transmitting hub for a LEAM that includes mechanical arrangement 20. More particularly, an electronics assembly 56, generally represented by dashed lines in FIG. 6, is retained in housing 22, and electronics assembly 56 is configured for electrically interconnecting light sources 54 (FIG. 7) to an external power source (not shown). Housing 22 can additionally contain sensory and communications devices, as discussed below in reference to FIG. 30. Housing 22 may include one, two, or more distinct compartments that may be defined by voltage classification. For example, alternating current (AC) line voltage devices may be placed at the top portion of housing 22, while lower AC or direct

current (DC) voltage devices may reside at the bottom portion of housing 22. Power may enter housing 22 from above and may then be distributed to the various devices within electronics assembly 56.

[0053] In some embodiments, an outer surface 58 of housing 22 includes a plurality of fins 60 extending into air gap 42 (FIG. 3) between housing 22 and heat dissipating structure 24. Fins 60 effectively increase a surface area of outer surface 58 of housing 22 to facilitate rapid cooling of the housed electronics assembly 56 to yield enhanced lifespan for the components of electronics assembly 56. Accordingly, the configuration of mechanical arrangement 20 enables cool device operation by the physical separation of electronics assembly 56 in housing 22 and light sources 54 within heat dissipating structure 24, and the free flow of air around both housing 22 and heat dissipating structure 24. Furthermore, housing 22 containing electronics assembly 56 is protected from moving objects in the immediate area by the surrounding heat dissipating structure 24.

[0054] Support arms 26 provide structural support for heat dissipating structure 24 while structurally isolating structure 24 from housing 22. In some embodiments, support arms 26 may have a generally V- or U-shaped cross sectional configuration, having a top removable cover 62 (see FIGS. 1 and 3), where removable cover 62 is wider than a base 63 (see FIG. 2) of each of support arms 26. The shape of support arms 26 induces free air flow upwardly around support arms 26, again to provide effective cooling. As generally shown in FIG. 3, at least one of support arms 26 includes an interior passage 64 (revealed when cover 62 is removed) for directing wiring 66 from electronics assembly 56 (FIG. 6) retained in housing 22 to each of lamp seats 50.

[0055] Support arms 26 may additionally provide structural support for other devices (discussed in connection with FIGS. 19-28). By way of example, an exterior surface 68 of cover 62 may additionally include a plug-in receptacle 70 and mounting holes 72 formed therein. A portion of wiring 66 may be routed to plug-in receptacle 70. Thus, external devices (not shown) may be removably mounted on exterior surface 68 of at least one of support arms 26. Such an external device can include a plug element 74 (see FIG. 19) that is attachable to plug-in receptacle 70 so that the external device has communication and power connectivity to electronics assembly 56 (FIG. 6).

[0056] Accordingly, mechanical arrangement 20 provides mechanical scalability. This scalability permits flexibility in choice of light output, a particular reflector assembly, and device choice and quantity, without having to re-design the form of mechanical arrangement 20. That is, housing 22, heat dissipating structure 24, and support arms 26 of mechanical arrangement 20 enable mechanical scaling thereby allowing for the same base architecture to be used in a variety of applications. These applications may include higher light output, different optical requirements, and device mix requirements.

[0057] Referring now to FIGS. 7-12, FIG. 7 shows a top perspective view of an integrated ceiling device, referred to as a Local Environmental Area Manager (LEAM), 80 in accordance with another embodiment FIG. 8 shows a bottom perspective view of LEAM 80. FIG. 9 shows a top view of LEAM 80. FIG. 10 shows a bottom view of LEAM 80. FIG. 11 shows a side view of LEAM 80, and FIG. 12 shows a side sectional view of LEAM 80. In general, LEAM 80 includes mechanical arrangement 20, electronics assembly 56 (generally represented in FIG. 12) retained in housing 22 of mechanical arrangement 20, and a refractor assembly 82 retained on heat dissipating structure 24 via a frame 84. In this example, frame 84 is secured to heat dissipating structure 24 via screws 86 attached to bores 47 (FIG. 1) in vertical walls 45 (FIG. 1) of heat dissipating structure 24. LEAM 80 may be adapted for use in a commercial environment where diffuse lighting, low glare, and an aesthetically pleasing appearance may be required. As such, along with refractor assembly 82, a reflector assembly 85 may be supported by support arms 26 and heat dissipating structure 24 in order to diffuse the light and/or to reduce glare from light sources 54.

[0058] As particularly shown in FIGS. 7, 9, 11, and 12, LEAM 80 includes a mounting cap 88 that couples to a top end 89 of housing 22 via fasteners 90. Mounting cap 88 may employ a

conventional power hook hanger **92**. Power hook hanger **92** provides a fastening means for coupling LEAM **80** to an exterior location, such as the ceiling of a building. Additionally, power hook hanger **92** is configured to enable the passage of wiring **93** (see FIG. **30**) so that LEAM **80** can be powered via building power.

[0059] Mounting cap **88** may additionally include provisions for data line connectivity via a data line receptacle **94** installed in mounting cap **88** and operatively connected to electronics assembly **56**. Data line receptacle **94** may be any receptacle suitable for data transfer such as, for example, an RJ45 receptacle, Universal Serial Bus (USB) receptacle, and the like. Data line receptacle **94** may be configured for attachment of a data line **96** (see FIG. **12**) between electronics assembly **56** and a remote device (not shown) to enable a transfer of data to and/or from electronics assembly **56**. Additionally, or alternatively, an antenna **98** may be installed in mounting cap **88**. Antenna **98** may be operatively connected to electronics assembly **56**. Antenna **98** may be configured for receiving and/or transmitting data between electronics assembly **56** and a remote device (not shown). Accordingly, implementation of data line receptacle **94** and/or antenna **98** enables communication between a remotely located control station or monitoring processor (not shown) and electronics assembly **56**.

[0060] As particularly shown in FIGS. **8**, **10**, and **12**, LEAM **80** further includes a removable access door **100** that couples to a bottom end **102** of housing **22**. Access door **100** can enable servicing of the devices that form electronic assembly **56** retained within housing **22** of heat dissipating structure **24**. Access door **100** may also incorporate one or more devices. For example, a speaker/microphone **104** and/or a smoke detector/air quality sensor **106** may be installed in access door **100**. Additionally, or alternatively, a camera/occupancy sensor **108** may be installed in access door **100**. Some embodiments may include multiple controllable devices that make up electronic assembly **56**. Accordingly, a series of switches **110** and/or indicator lights **112** may be installed in access door **100** in order to activate/deactivate and/or monitor the operation of the devices that make up electronic assembly **56**. Examples of the electronic components of light fixture **80** are discussed below in reference to FIG. **30**.

[0061] Now with particular reference to FIG. **12**, refractor assembly **82** is located at outer perimeter **44** of heat dissipating structure **24**. Reflector assembly **85** is located at inner perimeter **46** of heat dissipating structure **24**, and light sources **54** are positioned between refractor and reflector assemblies **82** and **85**, respectively. Refractor assembly **82** exhibits a first height **113** extending downwardly from a location **114**, i.e., the horizontal plane, of light sources **54**, and reflector assembly **85** exhibits a second height **115** extending downwardly from location **114** of light source **54**. In an embodiment, first height **113** is greater than second height **115**. More particularly, first height **113** of refractor assembly **82** may be at least one and one quarter times greater than second height **115** of reflector assembly **85**.

[0062] Together, refractor assembly **82** and reflector assembly **85** form an optical assembly **116** which is supported by, i.e., secured onto, heat dissipating structure **24**. Accordingly, refractor assembly **82** may be formed from a translucent glass, or some other translucent material. Furthermore, refractor assembly **82** may employ prismatic optics. In contrast, reflector assembly **82** may be formed from a highly reflective plastic, a material having a reflective material sputtered or otherwise deposited on it, or a polished metal. Additionally, reflector assembly **85** may employ segmented optics. In an embodiment reflector assembly **85** exhibits a profile, and in this configuration, an outwardly convex profile that is configured to redirect light emitted from light source **54** toward refractor assembly **85**, as well as to downwardly direct light emitted from light source **54**.

[0063] Optical assembly **116**, including refractor assembly **82** and reflector assembly **85**, functions to effectively redirect light from light sources **54** in order to improve light source uniformity, to increase a “glow effect”, and to reduce glare. Such a structure may obtain optical efficiencies of greater than ninety-five percent. Additionally, the difference between heights **113** and **115** largely

prevents direct visibility of light sources **54** over sixty degrees from nadir, where the nadir (in accordance with the Illuminating Engineering Society of North America) is defined as the angle that points directly downward, or zero degrees, from a luminaire. Accordingly, FIG. **12** shows a nadir as corresponding to a longitudinal axis **118** of LEAM **80**. As further shown in FIG. **12**, an approximately sixty degree angle is formed between longitudinal axis **117**, i.e., the nadir, and a virtual line intersecting the bottommost edges of refractor assembly **82** and reflector assembly **85**. It is known that light emitted in the eighty degree to ninety degree zone from nadir is more likely to contribute to glare. Accordingly, the difference between heights **113** and **115** effectively limits the potential for glare by directing the light within the sixty degree from nadir range.

[0064] Referring to FIGS. **13-18**, FIG. **13** shows a top perspective view of an integrated ceiling device, referred to as a Local Environmental Area Manager (LEAM), **120** in accordance with another embodiment. FIG. **14** shows a bottom perspective view of LEAM **120**. FIG. **15** shows a top view of LEAM **120**. FIG. **16** shows a bottom view of LEAM **120**. FIG. **17** shows a side view of LEAM **120**, and FIG. **18** shows a side sectional view of LEAM **120**. In general, LEAM **120** includes mechanical arrangement **20**, electronics assembly **56** (generally represented in FIG. **18**) retained in housing **22** of mechanical arrangement **20**, and frame **84** secured to heat dissipating structure **24**, as described above.

[0065] LEAM **120** may not include refractor assembly **82** and reflector assembly **85**, as discussed in connection with LEAM **80** of FIG. **7**. Rather, only frame **84** may be present to provide some amount of protection for light sources **54** within LEAM **120** from movable objects in the location at which LEAM **120** resides. LEAM **120** may be adapted for use in an industrial environment where high brightness and a relatively narrow beam pattern may be called for. In some configurations, LEAM **120** may include an optical assembly, supported by heat dissipating structure **24**, in the form of a plurality of individual reflector assemblies **122**. As such, each light source **54** is surrounded by an individual one of reflector assemblies **122**. Reflector assemblies **122** may be supported or retained by bottom face **48** of heat dissipating structure **24** and support arms **26** in order to focus the light pattern from light sources **54**. In some embodiments, individual reflector assemblies **122** may have different optical properties. Additionally, light sources **54** may have varying light output. Thus, a combination of reflector assemblies **122** and light sources **54** can be selected to provide a desired lighting pattern.

[0066] Like LEAM **80** (FIG. **7**), LEAM **120** includes mounting cap **88** having power hook hanger **92**, data line receptacle **94**, and/or antenna **98** installed therein. Additionally, LEAM **120** includes access door **100** having speaker/microphone **104**, smoke detector/air quality sensor **106**, camera/occupancy sensor **108**, switches **110** and/or indicator lights **112** incorporated therein as discussed above.

[0067] Referring to FIGS. **19-23**, FIG. **19** shows a bottom perspective view of a device **130** that may be mounted to mechanical arrangement **20** (FIG. **1**) in accordance with another embodiment. FIG. **20** shows a top perspective view of device **130**. FIG. **21** shows a side view of device **130**. FIG. **22** shows a bottom view of device **130**, and FIG. **23** shows a top view of device **130**. In an embodiment, device **130** may be an uninterruptible power supply (UPS) that can provide emergency power when the input power source, in this case mains power, fails. Accordingly, device **130** will be referred to hereinafter as UPS **130**.

[0068] UPS **130** includes a housing **132** and the necessary electronics (not shown) retained in housing **132** for supplying emergency power when incoming voltage falls below a predetermined level. The electronics may include, for example, a charger, backup battery, and DC-AC input inverter (not shown) as known to those skilled in the art. As shown in FIGS. **22** and **23**, housing **132** has a profile that is adapted to interface with outer surface **58** (FIG. **1**) of housing **22** (FIG. **1**) of mechanical arrangement **20** (FIG. **1**). A bottom surface **134** of housing **132** includes plug element **74** that is attachable to plug-in receptacle **70** (FIG. **3**) formed in cover **62** (FIG. **3**) of one of support arms **26** (FIG. **3**) so that UPS **130** is electrically connected to electronics assembly **56** (FIG.

6), as discussed previously. Additionally, bottom surface **134** may include mounting holes **136** that mate with mounting holes **72** (FIG. 3) in cover **62**. Conventional fasteners (not shown) may be utilized to fasten housing **132** to cover **62** via mounting holes **136** and mounting holes **72**.

[0069] Referring to FIGS. 24-28, FIG. 24 shows a bottom perspective view of a device **140** that may be mounted to mechanical arrangement **20** (FIG. 1) in accordance with another embodiment. FIG. 25 shows a top perspective view of device **140**. FIG. 26 shows a side view of device **140**. FIG. 27 shows a bottom view of device **140**, and FIG. 28 shows a top view of device **140**. In an embodiment, device **140** may be an uplight that is positioned to cast its light in a direction opposite, for example, upwards, from the light cast by light sources **54** (FIG. 8). As such, device **140** is referred to hereinafter as uplight **140**. Uplight **140** may be activated alone or along with light sources **54** during normal operation when it is desirable to cast light upwards to provide all-round indirect illumination. Alternatively, or additionally, uplight **140** may be activated during an extended loss of mains power in order to provide emergency lighting.

[0070] Uplight **140** includes a housing **142** having a light source **144** installed in a top surface **146** of housing **142**. Electronics (not shown) may be retained in housing **142** for operating light source **144**, as known to those skilled in the art. In some configurations, light source **144** may be an LED or any other suitable light source. Accordingly, housing **142** may include a heat sink region **148** formed in one or more side walls **150** of housing **142**. Heat sink region **148** may include multiple fins that are configured to conduct the heat generated by light source **144** away from light source **144**.

[0071] As shown in FIGS. 27 and 28, housing **142** has a profile that is adapted to interface with outer surface **58** (FIG. 1) of housing **22** (FIG. 1) of mechanical arrangement **20** (FIG. 1). A bottom surface **152** of housing **142** includes another plug element **74** that is attachable to plug-in receptacle **70** (FIG. 3) formed in cover **62** (FIG. 3) of one of support arms **26** (FIG. 3) so that uplight **140** can be electrically connected to electronics assembly **56** (FIG. 6), as discussed previously. Additionally, bottom surface **152** may include mounting holes **154** that mate with mounting holes **72** (FIG. 3) in cover **62**. Conventional fasteners (not shown) may be utilized to fasten housing **142** to cover **62** via mounting holes **152** and mounting holes **72**.

[0072] FIG. 29 shows a side view of light fixture **80** of FIG. 13 with UPS **130** and uplight **140** retained on light fixture **80**. As discussed above, each of UPS **130** and uplight **140** are removably mounted to an exterior surface of one of support arms **26** (FIG. 3). More particularly, each of UPS **130** and uplight **140** is mounted to top removable cover **62** (FIG. 3) of one of support arms **26** and abuts housing **22** of mechanical arrangement **20**. Additionally, plug element **74** (FIGS. 19 and 24) of its respective UPS **130** and uplight **140** is attached with its respective plug-in receptacle **70** (FIG. 3) installed in top cover **62** so as to electrically connect UPS **130** and uplight **140** to electronics assembly **56** (FIG. 6) retained in housing **22**.

[0073] FIG. 30 shows a block diagram of electronics assembly **56** including a variety of devices that may be included within electronics assembly **56** of any of LEAM **80** (FIG. 7), LEAM **120** (FIG. 13), or any of a number of LEAM designs. In general electronics assembly **56** includes sufficient processing power coupled with sensor perception in order to emulate the human capacity of make actionable, predictable, and accurate decisions in response to environmental changes. To that end, electronics assembly **56** is capable of accepting and operating a variety of devices independently or in unison. These devices may be miniaturized to provide a broader platform for a larger number of devices with greater interactive capabilities. Thus, electronics assembly **56** may be considered a Local Environment Area Manager (LEAM) where the lighting related components provide the physical platform for the LEAM. By way of explanation, electronics assembly **56** is described in connection with LEAM **80**. As such, reference should be made to FIGS. 7-12 concurrent with the following description.

[0074] Components of electronics assembly **56** include, but are not limited to, one or more power supplies **160**, a communicator element **162**, one or more processors **164**, and an array of devices

166 capable of data/signal input and output, each of which may be suitably connected via a power bus **168** (represented by solid lines) and a data bus **170** (represented by dotted lines). [0075] In general, power supply **160** receives line power **172** via wiring **93** routed through power hook hanger **92** and converts line power **172** to the power needed to operate the various devices of electronics assembly **56**. Power supply **160** may be modular and scalable having one or more input power channels **174** and output power channels **176**. Input and output power channels **174**, **176** may be programmable with flexibility to change the power supplied and device-specific power operational parameters as needed. Power supply **160** may have an optional dedicated processor **178**, represented in dashed line form, governing the power from power supply **160** while maintaining real-time communication with processor **164** of electronics assembly **56**. In some embodiments, power supply **160** may also have direct communication capability with an external network (not shown).

[0076] Data output from power supply **160** may include reporting on the quality of the input power, the operational temperature of power supply **160**, the power consumption of power supply **160** including client devices such as communicator element **162**, processor **164**, and array of devices **166**, time of usage broken down by device, and operational anomalies. Power supply **160** processes the highest electrical load of electronics assembly **56**. Therefore, power supply **160** may be located in the upper region or an upper compartment of housing **22**. The upper region of housing **22** has three hundred and sixty horizontal degrees of exposure to cooling air circulation and full exposure to cooling air at mounting cap **88** for providing effective cooling to the housed power supply **160**. The circuit boards (not shown) for power supply **160** may be wired by a conventional method or engaged by plug-in connectors. Additionally, the circuit boards may be encased or open and may be secured directly to housing **22** by mounting them along the inner perimeter of housing **22**.

[0077] As a local environment area manager, electronics assembly **56** includes communicator element **162** in order to permit direct or via processor **164** communication with onboard array of devices **166**. Additionally, communicator element **162** may be configured to enable communication between a plurality of LEAMs **80**, to enable communication with a local or remote building management system, and/or with local or remote clients. Such clients could be corporate offices or first responders needing real time input about a specific location in a building. Communicator element **162** may employ radio frequency (RF) communication via antenna **98**, power line communication (PLC) carrying data on the mains power line carrying line power **172**, a dedicated data line such as data line **96** connected with data line receptacle **94**, or any combination thereof.

[0078] In some embodiments, each LEAM **80** may be initialized with a unique address and an option. ability to assign a sub-address to all devices within LEAM **80**. In this manner, the operational integrity of the various elements of electronics assembly **56** may be monitored and any anomalies with onboard devices may be alerted, identifying the nature of the anomaly and possible recommendations for action.

[0079] Processor **164** can contain resident memory **180** that may be programmed with control code **182** prior to delivery to a building, during commissioning, or at any time thereafter. Programming may be performed by a wired connection to a port, e.g., data line **96** connected to data line receptacle **94** or wirelessly via antenna **98**. System updates and device specific updates to control code **182** may occasionally be performed with occasional device upgrades.

[0080] Processor **164**, executing control code **182**, may be configured to receive local device sensory input from one or more devices of array **166**, and then compile this information in accordance with pre-programmed instructions. Processed information may then be converted to actionable output to array of devices **166**. In addition, processor **164** may communicate with neighboring or remote devices and may transmit instructions, instructions and data, or instructions, data and images. Processing power may vary among electronic assemblies **56** of a variety of LEAMs, based on the application's specific needs.

[0081] Control code **182** may be multi-device relational software designed to operate array of

devices **166** in unison. Control code **182** may be scalable by modules, where each module relates to the functionality of an associated device and its relation to other onboard devices and the entire network's devices. Control code **182** may be provided with input tables such as schedules and set points, as well as alert parameters and operational reports. In addition, control code **182** can be customized for specific applications and may include self-learning modules. Processor **164** has sufficient memory **180** associated therewith in order to access and act on pertinent information in real time. Additionally, control code **182** may be provided with a self-reporting module associated with each device in array **166** in order to report the device's operational condition and provide alerts when the device performs outside its optimal performance range.

[0082] Array of devices **166** includes light sources **54** and uplight **140**, and may include one or all of the following: camera **108**; a compass **186**; speaker/microphone **104** or a combination thereof; smoke detector/air quality sensor **106**; an occupancy sensor **192**; a thermostat **194**; a backup battery, e.g., UPS **130** (FIG. **19**); and noise suppression circuitry **196**. The devices presented in array **166** should not be considered to be all inclusive. That is, the devices represented in array **166** may additionally include other building system and monitoring devices and circuits not listed herein.

[0083] The various devices within array **166** may be utilized in connection with comfort control, life safety, loss prevention, marketing analysis, asset management functions, and/or operational optimization. Comfort control may entail lighting uniformity, sound control including suppression, temperature control, air quality control, and so forth. Life safety may entail air quality monitoring, local and remote alarming, lighting and sound delineation of egress pathway, live feed to first responders, the identification of "hot spots" in event of fire or crime, and so forth. Loss prevention may entail system alarming for operational anomalies, theft prevention via behavioral software analysis, produce spoilage avoidance by monitoring local ambient air temperature, and so forth. Marketing analysis may entail monitoring or customer traffic pattern, customer behavior, customer gender, cash register wait time, customer volume, customer time of day week and month visit, customer demographics, and so forth. Asset management may entail space utilization reporting, facility design performance analysis, asset inventory control, asset depreciation record, and so forth. Operational optimization may entail energy usage monitoring, reduction in maintenance men hours via event and alarms reporting, event recordation, product and system performance evaluation, and so forth.

[0084] One feature of LEAMs **80, 120** having mechanical arrangement **20** is the optimization of exposure to the surrounding cooling air and airflow at least two hundred and seventy degrees across the vertical axis of heat dissipating structure **24**, as well as the vertical and horizontal axes of housing **22**. The form of mechanical arrangement **20** may be guided by its components, such as heat dissipating structure **24** at the perimeter of mechanical arrangement **20**, housing **22** at the center of mechanical arrangement **20**, and support arms **26** bridging between structure **24** and housing **22** through which power flows to light sources **54** in heat dissipating structure **24**. The form of mechanical arrangement **20** may enable superior thermal heat management capabilities of LEAMs **80, 120**.

[0085] Another feature of LEAMs **80, 120** having mechanical arrangement **20** and electronics assembly **56** is the capability to mount lighting and non-lighting related environment sensory devices (e.g., speaker/microphone **104**, smoke detector/air quality sensor **106**, and camera/occupancy sensor **108**) in access door **100** at bottom end **102** of housing **22**, as well as mounting communication devices (e.g., data line receptacle **94** and antenna **98**) in mounting cap **88** at top end **89** of housing **22**. With the proximity of communication devices to the power entry for LEAMs **80, 120** at power hook hanger **92**, efficiencies are achieved due to centralized placement location for the lighting and non-lighting related devices. Moreover, the separation between the housed electronics assembly **56** and light sources **54** enables effective cooling of the housed electronics assembly **56**, as well as protection from exterior forces.

[0086] Another feature of LEAMs **80, 120** with centrally located housing **22** is the capability to access power supply **160** at bottom end **102** of housing **22** via removably mounted access door **100**. Certain components, such as power supply **160** may be readily and quickly inserted and removed through guiding slots on an inner perimeter of housing **22**. Since access door **100** is removable, all devices within housing **22** may be equipped with a quick disconnect in order to install, remove, and replace any of the devices within housing **22**.

[0087] Yet another feature of LEAMs **80, 120** includes the capability to provide backup emergency battery, e.g., UPS **130**, whose power may be selectively distributed to all essential services and devices during an emergency, which can be received either from a light fixture driver or a secondary step-down power device. In some embodiments, UPS **130** may be connected to the light fixture audio system, e.g., speaker/microphone **104**. Additionally, UPS **130** may be networked with other input/output onboard environmental data collection, assessment, and operational devices, and have remote communication capability.

[0088] Another feature of LEAMs **80, 120** includes an emergency light system, e.g., light sources **54**, that may be used to delineate an egress path by supplementing the ambient light level to identify the directionality of the path to egress doors. This down light feature, using light sources **54** may be supplemental with a strobe light. In some embodiments, speaker/microphone **104** may be used to broadcast a pathway direction identifier.

[0089] Another feature of LEAMs **80, 120** is the capability to operate one or several onboard devices from array **166**, such as backup battery **130**, speaker/microphone **104**, smoke detector/air quality sensor **106**, camera/occupancy sensor **108**, communicator element **162**, and compass **186** in unison, based on real time information processed and programmed instructions.

[0090] A feature of LEAMs **80, 120** having electronics assembly **56** entails the capability to perform auto-commissioning of a network of light fixtures **80**. For example, each of LEAMs **80, 120** includes a discrete address, camera **108**, communicator element **162**, processor **164** and/or remote processors. Processor **164** and/or the remote processors may include an electronic map showing each of LEAMs **80, 120** by its associated discrete address and its relative location to the entire network of LEAMs **80, 120**. Auto-commissioning commences following association of one LEAMs **80, 120** with its corresponding placement on the electronic map.

[0091] Another feature of LEAMs **80, 120** having electronics assembly **56** entails light control at its local location. For example, each of LEAMs **80, 120** includes a discrete address, camera **108** with an integrated light meter, compass **186**, communicator element **162**, processor **164** and/or remote processors. Processor **164** and/or the remote processors may maintain a predetermined light level by dimming or turning LEAMs **80, 120** on or off through processing in real time local zone illumination conditions data obtained by camera **108** and preprogrammed local or remote controller instructions.

[0092] Another feature of LEAMs **80, 120** having electronics assembly **56** entails the optimization of local and entire space environmental conditions. Optimization methodology may utilize data from camera **108**, occupancy sensor **192**, as well as other onboard sensor devices such as processor **164**, thermostat **194**, communicator element **162**, processor **164** and/or remote processors to process data and act in real time on changing conditions while operating within programmatic instruction guidelines.

[0093] Another feature of LEAMs **80, 120** having electronics assembly **56** entails the capability to collect environmental conditions data via camera **108** and relay the data to local processor **164** and/or remote processors. The data collected by camera **108** may include, but is not limited to, the functionality of devices such as in occupancy sensing, a light meter output, a traffic count, human load density analysis, time of day activity logging, and photographic and thermal imagery. The processed data obtained by camera **108** with or without additional information processed from other non-camera devices within LEAMs **80, 120** facilitate optimal operation of LEAMs **80, 120**.

[0094] Another feature of LEAMs **80, 120** having electronics assembly **56** entails the capability to

function as a public announcement, sound, and alarming system through the provision of audio input/output via microphone/speaker **104**. Additionally, microphone/speaker **104** may be networked with other input/output onboard environment data collection, assessment, and operational devices, and have remote communication capability.

[0095] Another feature of LEAMs **80, 120** having electronics assembly **56** entails the implementation of smoke detector/air quality sensor **106**. Smoke detector/air quality sensor **106** may also be networked with other input/output onboard environmental data collection, assessment, and operational devices, and have remote communication capability.

[0096] In summary, embodiments described above resolve a number of the mechanical, thermal, electrical, and architectural challenges that are commonly associated with integrated ceiling system devices and particularly with high-output LED light fixture design. Furthermore, the structural configuration of the LEAM makes the LEAM suitable for use in a wide variety of environments, such as, commercial, institutional, and industrial applications. Additionally, the LEAM including the mechanical arrangement and electronics assembly may assume partial or full control over the ambient environment in the vicinity of the LEAM, integrating operational logic traditionally associated with isolated disciplines' networks of heating, ventilation, and air conditioning (HVAC) monitoring devices, fire protection devices, air quality monitoring devices, input/output audio devices, temperature and humidity devices, security and normal operation monitoring cameras, occupancy sensors, lighting controls, and so forth. Consequently, the LEAM including the mechanical arrangement and the electronics assembly yields significant improvements in terms of the integration of a variety of building system functions combined with the quintessential need for suitable ambient lighting in an aesthetically pleasing form factor.

[0097] While the principles of the inventive subject matter have been described above in connection with specific apparatus configurations described above, it is to be clearly understood that this description is made only by way of example and not as a limitation on the scope of the inventive subject matter. For example, embodiments may be implemented in systems having other architectures as well. The various functions or processing blocks discussed herein and illustrated in the Figures may be implemented in hardware, firmware, software or any combination thereof. Further, the phraseology or terminology employed herein is for the purpose of description and not of limitation.

[0098] The foregoing description of specific embodiment reveals the general nature of the inventive subject matter sufficiently so that others can, by applying current knowledge, readily modify and/or adapt it for various applications without departing from the general concept. Therefore, such adaptations and modifications are within the meaning and range of equivalents of the disclosed embodiments. The inventive subject matter embraces all such alternatives, modifications, equivalents, and variations as fall within the spirit and broad scope of the appended claims.

Claims

1. An integrated ceiling device comprising: a heat dissipating structure comprising a unitary heat sink coupled to a plurality of fins disposed on top of the heat sink and distributed around a perimeter of the heat sink, and a protective ring that couples to the outer perimeter of the heat dissipating structure, an end of at least a subset of the plurality of fins extend out toward the protective ring; an electronics housing sized to retain an electronics assembly therein, at least a portion of the electronics housing is disposed above a central portion of the heat dissipating structure and has a vertical axis that is substantially aligned vertical axes of a central through opening and the heat dissipating structure, a gap between the heat dissipating structure and the electronics housing enables the flow of air through the central opening to above the electronics housing, wherein a plurality of second through openings form a through air gap between the

protective ring and an outer perimeter of the heat dissipating structure, the central through opening being radially closer to the vertical axis of the heat dissipating structure than the plurality of second through openings, the electronics housing is enclosed and coupled to the heat dissipating structure, the heat dissipating structure is electrically coupled to the electronics housing and at least one light source couples to a bottom of the heat sink, and respective exterior surfaces of the electronics housing and the heat dissipating structure are exposed to allow air flow.

2. The device of claim 1, wherein an optical lens is coupled to the heat dissipating structure.

3. The device of claim 1, wherein light emitted by at least one lamp that is coupled to the bottom surface of the heat dissipating structure is directed toward a floor below the integrated ceiling device.

4. The device of claim 1, wherein a diameter of the heat dissipating structure is larger than a diameter of the electronics device housing.

5. The device of claim 1, further comprising at least one mechanical arm that couples the electronics housing to the heat dissipating structure.

6. The device of claim 1, wherein air is free to flow between the heat dissipating structure and the electronics device housing through the central opening in the heat dissipating structure and across the at least one mechanical arm.

7. The device of claim 1, wherein the plurality of fins extend into the through air gap between the protective ring and the outer perimeter of the heat dissipating structure.

8. The device of claim 1, further comprising a detachable top cap disposed above the heat dissipating structure.

9. An integrated ceiling device comprising: an electronics housing configured to retain an electronics assembly therein; a heat dissipating structure comprising a unitary heat sink coupled to a plurality of fins disposed on top of the heat sink and around a perimeter of the heat sink; a light source; and at least one mechanical arm extends between the electronics housing and the heat dissipating structure to form a through air gap therebetween, wherein the electronics housing supports a weight of the integrated ceiling device, is at least partially disposed above the heat dissipating structure, is positioned to have a central axis that is substantially aligned with a vertical axis of a central through opening, and a vertical axis of the heat dissipating structure, and is enclosed, wherein the heat dissipating structure is electrically coupled to the electronics housing and the light source is coupled to a bottom of the heat sink, and under a condition that the light source is energized, air that rises from below the integrated ceiling device flows through, and then above, the central opening in the heat dissipating structure to cool the heat dissipating structure and the electronics housing.

10. The device of claim 9, further comprising a protecting ring that is disposed around a periphery of the heat dissipating structure so as to protect the integrated device from being damaged by moving objects that strike the integrated ceiling device.

11. The device of claim 9, wherein air is free to flow through one or more air gaps between the protective ring and an outer perimeter of the heat sink.

12. The device of claim 9, wherein the heat dissipating structure is positioned to protect the electronics housing from being directly impacted by moving objects.

13. The device of claim 9, further comprising a mounting hook coupled to the electronics housing that supports a weight of the integrated ceiling device.

14. The device of claim 9, wherein at least one of a refractor and a reflector is coupled to a bottom surface of the heat dissipating structure and directs light toward a floor below.

15. The device of claim 9, wherein the electronics housing has a cylindrical shape.

16. The device of claim 9, wherein an architecture of the integrated ceiling device is configured to accommodate an changeable level of output light via scaling up or down of the integrated ceiling device.

17. An integrated ceiling device comprising: a heat dissipating structure comprising a unitary heat

sink coupled to a plurality of fins disposed on top of the heat sink and distributed around a perimeter of the heat sink and a protective ring; an electronics housing sized to retain an electronics assembly therein, at least a portion of the electronics housing is disposed above a central portion of the heat dissipating structure and has a vertical axis that is substantially aligned with a vertical axis of the heat dissipating structure, a plurality of through openings form a through air gap between the protective ring and an outer perimeter of the heat dissipating structure, wherein the electronics housing is enclosed and coupled to the heat dissipating structure, a gap between the electronics housing and the heat dissipating structure exposes a substantial portion of the external surfaces of the electronics housing and the heat dissipating structure to flow of air, the heat dissipating structure is electrically coupled to the electronics housing, the integrated ceiling device further comprising at least one light source coupled to a bottom of the heat sink; a communication device is coupled to the electronics housing; and a power consuming device coupled to the integrated ceiling device and that is communicatively coupled to a remote power consuming device.

18. The device of claim 17, wherein the communication device is configured to perform at least one of wired communication or wireless communications.

19. The device of claim 17, further comprising an antenna that is electrically coupled to the electronics housing.

20. The device of claim 17, wherein the at least one light source is controllable by a remote controller.

21. The device of claim 17, further comprising a switch coupled to the electronics housing.

22. The device of claim 17, further comprising at least one power consuming device coupled to the integrated ceiling device that is controlled by a switch.

23. The device of claim 17, wherein the electronics housing includes a data conductor port that is different than a port to accommodate a power conductor.

24. The device of claim 17, further comprising a sensor that is electrically coupled to the electronics housing.

25. The device of claim 24, wherein the sensor is an occupancy sensor mounted to a floor facing surface of the integrated ceiling device.

26. The device of claim 17, wherein a first power source coupled electrically to the electronics housing is at a different voltage than a house power voltage applied to the electronics housing.

27. The device of claim 17, further comprising a power converter electrically coupled to the electronics housing, and which is configured to convert from AC to DC.

28. The device of claim 17, further comprising a processor coupled to the electronics housing.

29. The device of claim 17, further comprising a back-up power supply electrically coupled to the electronics housing.

30. The device of claim 17, further comprising a low voltage device coupled to the electronics housing and positioned below the electronics housing.
